

5ME42A(ME) – Manufacturing Technology

5th Semester B.E. Mechanical Engineering

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NC, CNC machines and Programming

Introduction to NC and CNC machines

CNC controls and programming

History of CNC

1949

US Air Force asks MIT to develop a "numerically controlled" machine.

1952

Prototype NC machine demonstrated (punched tape input)

1980-

CNC machines (computer used to link directly to controller)

1990-

DNC: external computer “drip feeds” control programmer to machine tool controller

Motivation and uses

To manufacture complex curved geometries in 2D or 3D was extremely expensive by mechanical means (which usually would require complex jigs to control the cutter motions)

Machining components with repeatable accuracy

Unmanned machining operations

Advantages of CNC

- Easier to program;
- Easy storage of existing programs;
- Easy to change a program
- Avoids human errors
- NC machines are safer to operate
- Complex geometry is produced as cheaply as simple ones
- Usually generates closer tolerances than manual machines

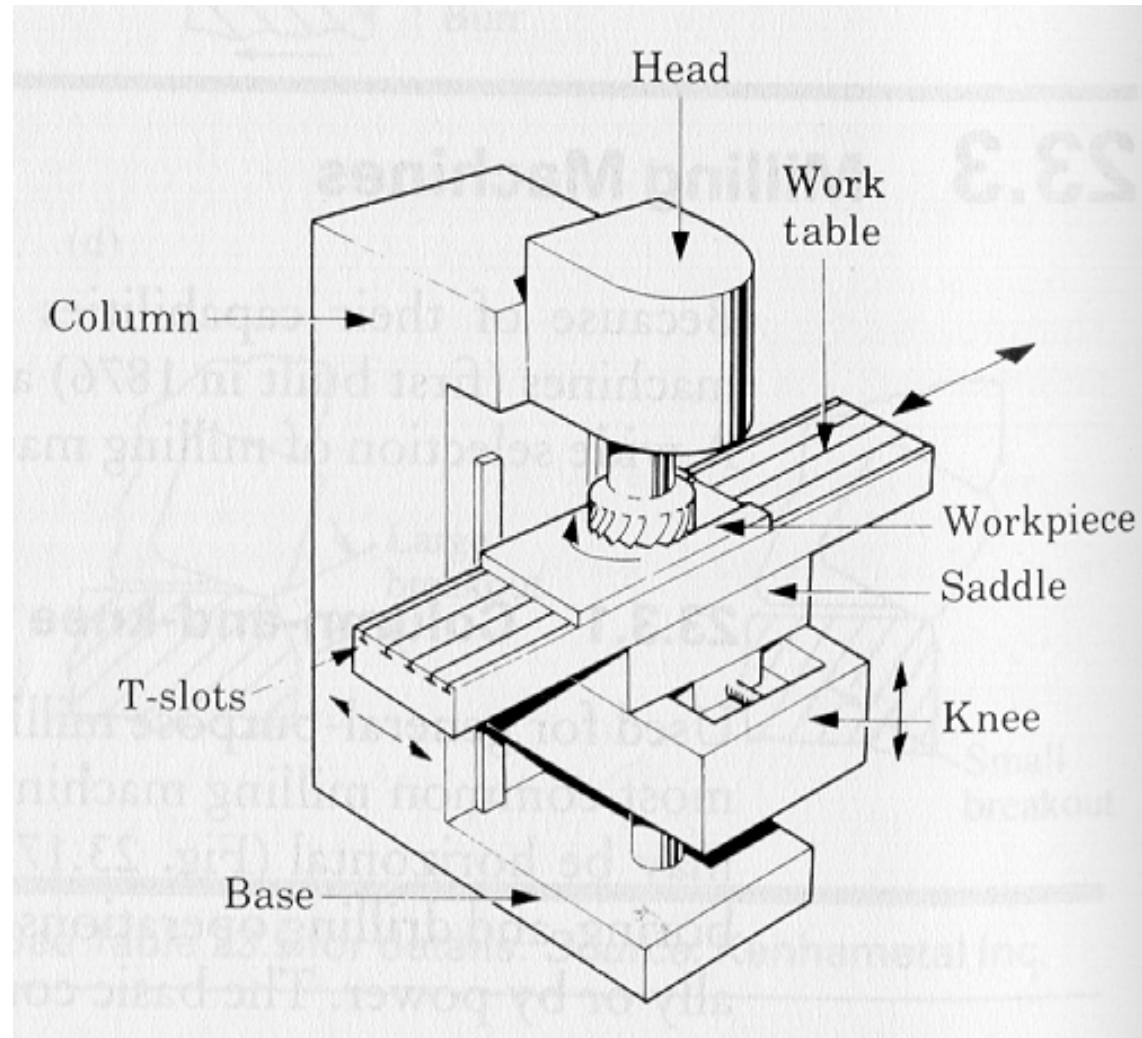
Conventional milling machines



Vertical milling machine

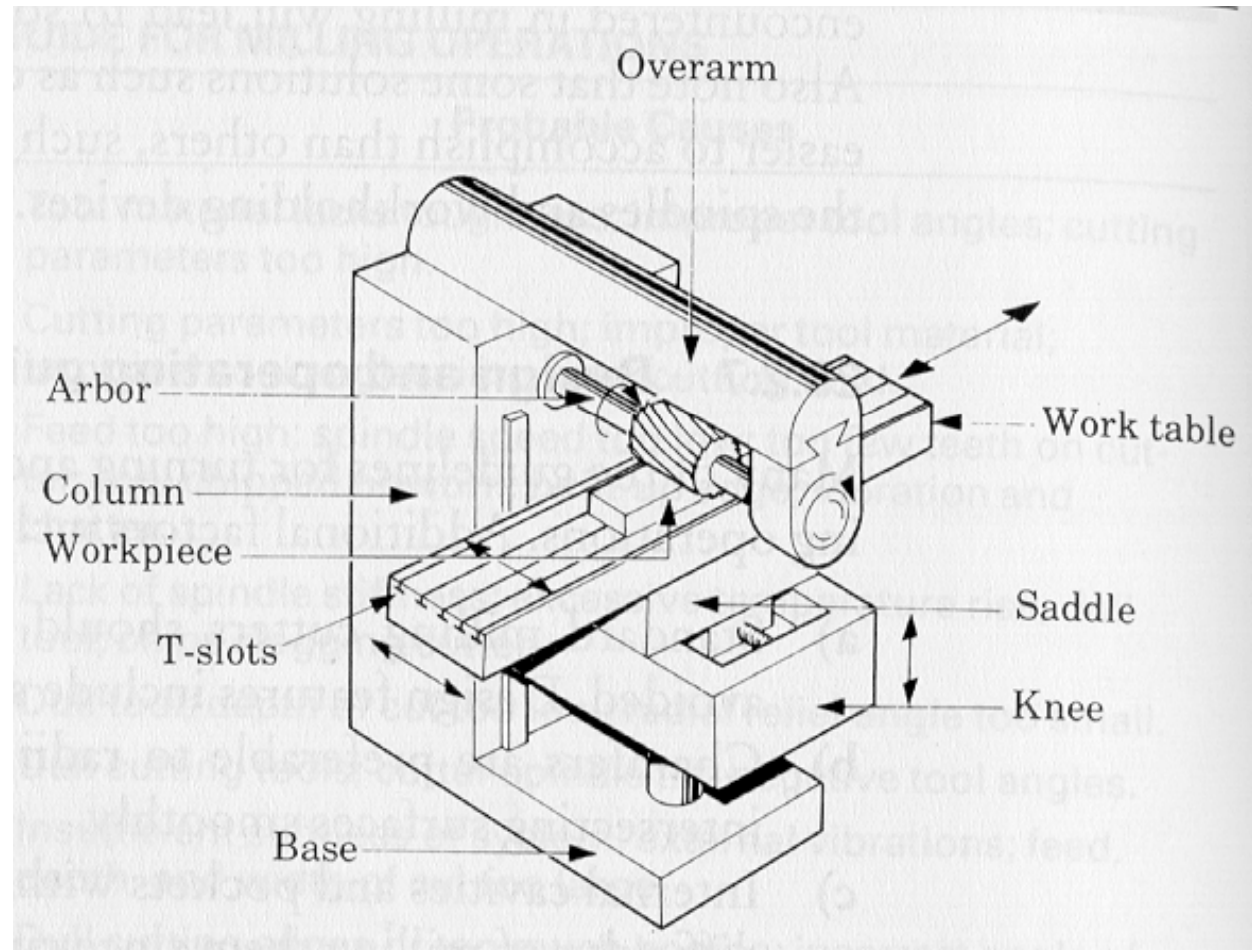
Conventional milling machines

Vertical Milling machine architecture



Conventional milling machines

Horizontal Milling machine architecture



How does the table move along X- Y- and Z- axes ?

CNC terminology

BLU: basic length unit →
smallest programmable move of each axis.

Controller: (Machine Control Unit, MCU) →
Electronic and computerized interface between operator and m/c

Controller components:

1. Data Processing Unit (DPU)
2. Control-Loops Unit (CLU)

Types of CNC machines

Based on Motion Type:

Point-to-Point or Continuous path

Based on Control Loops:

Open loop or Closed loop

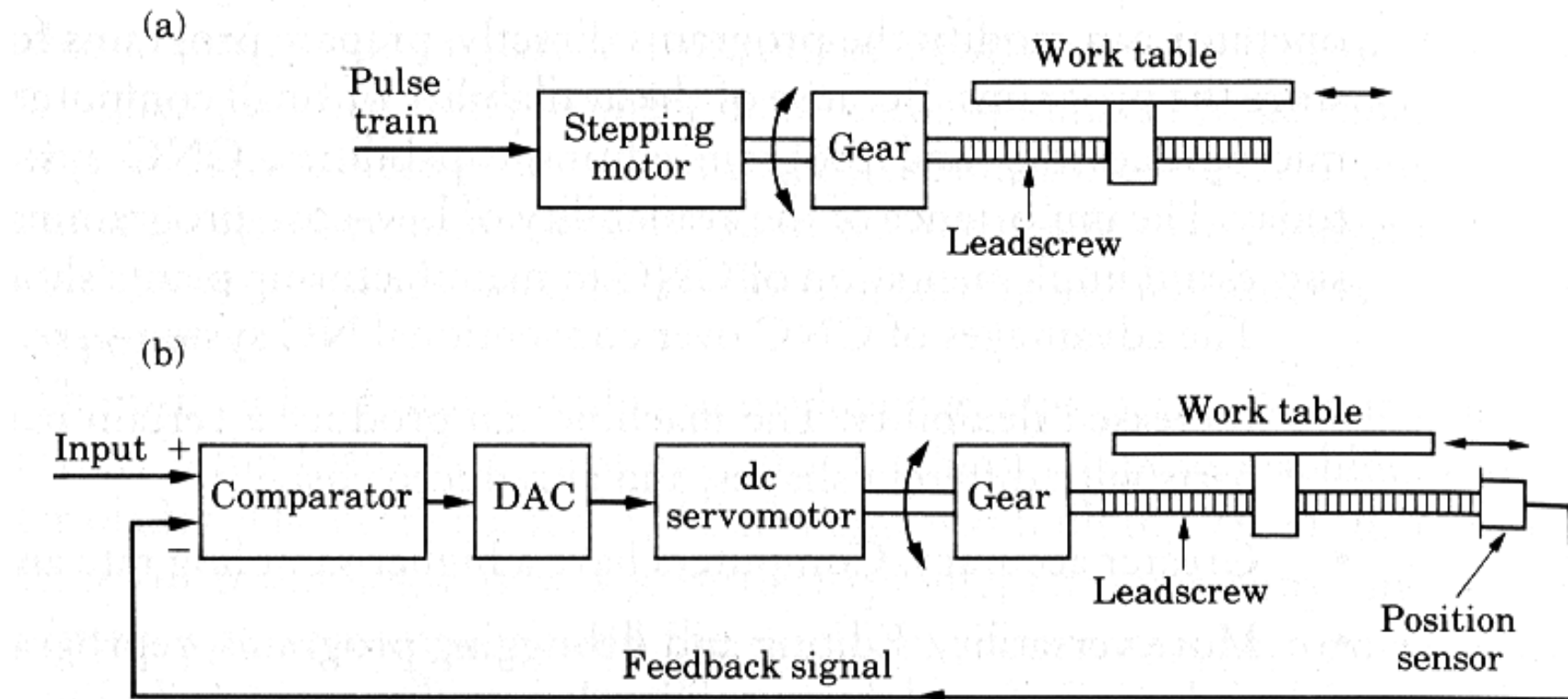
Based on Power Supply:

Electric or Hydraulic or Pneumatic

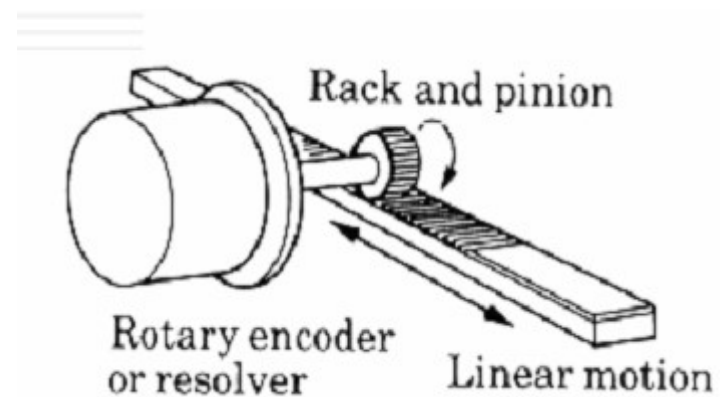
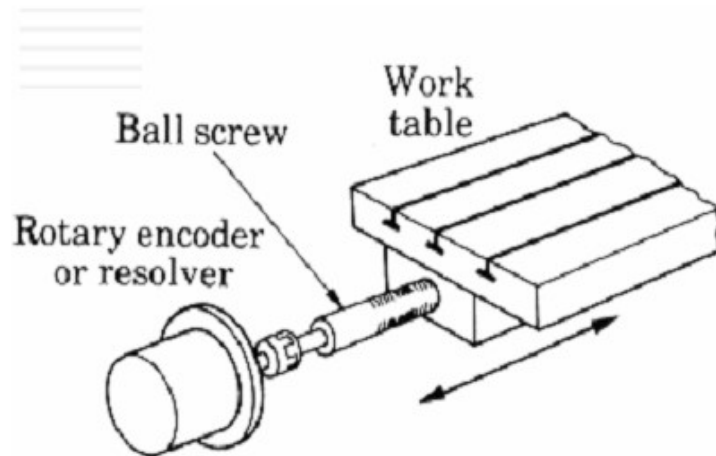
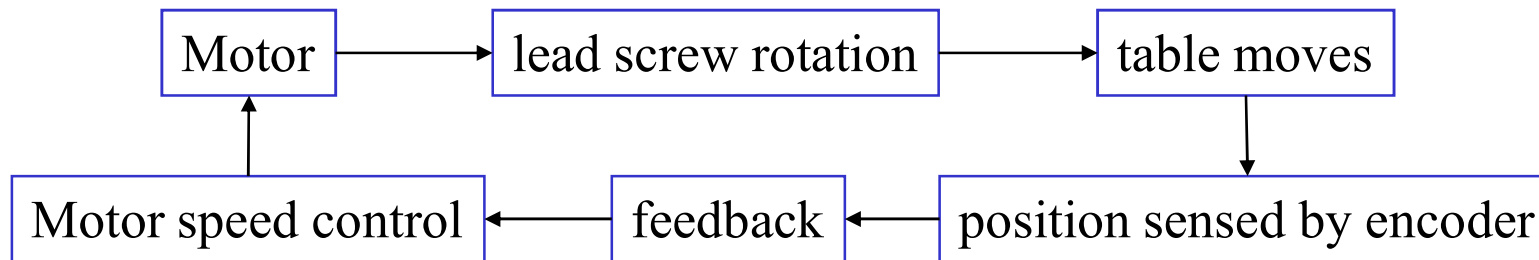
Based on Positioning System

Incremental or Absolute

Open Loop vs. Closed Loop controls



Components of Servo-motor controlled CNC



Two types of encoder configurations

Motion Control and feedback

Encoder outputs: electrical pulses (e.g. 500 pulses per revolution)

Rotation of the motor → linear motion of the table: by the **leadscrew**

The **pitch** of the leadscrew: horizontal distance between successive threads

One thread in a screw → **single start screw**: Dist moved in 1 rev = pitch

Two threads in screw → **double start screw**: Dist moved in 1 rev = $2 \times$ pitch

Manual NC programming

Part program: A computer program to specify

- Which tool should be loaded on the machine spindle;
- What are the cutting conditions (speed, feed, coolant ON/OFF etc)
- The start point and end point of a motion segment
- how to move the tool with respect to the machine.

Standard Part programming language: RS 274-D (Gerber, GN-code)

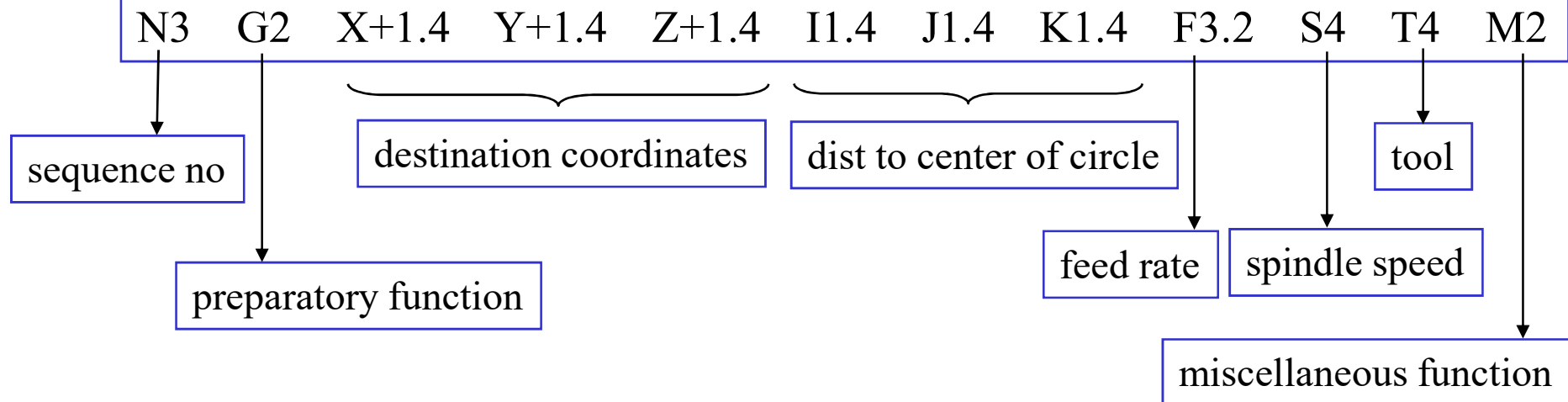
History of CNC

The RS274-D is a **word address format**

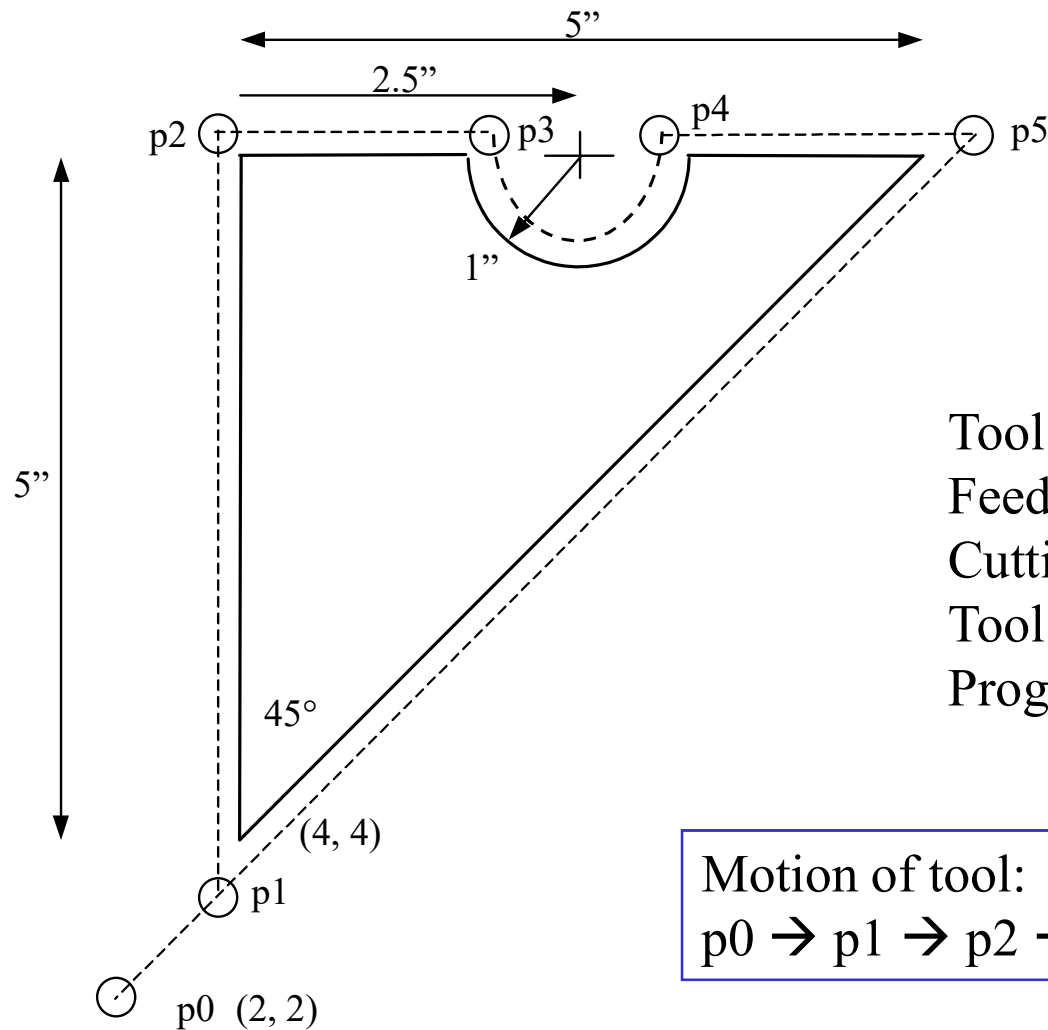
Each line of program == 1 **block**

Each block is composed of several instructions, or (**words**)

Sequence and format of words:



Manual Part Programming Example

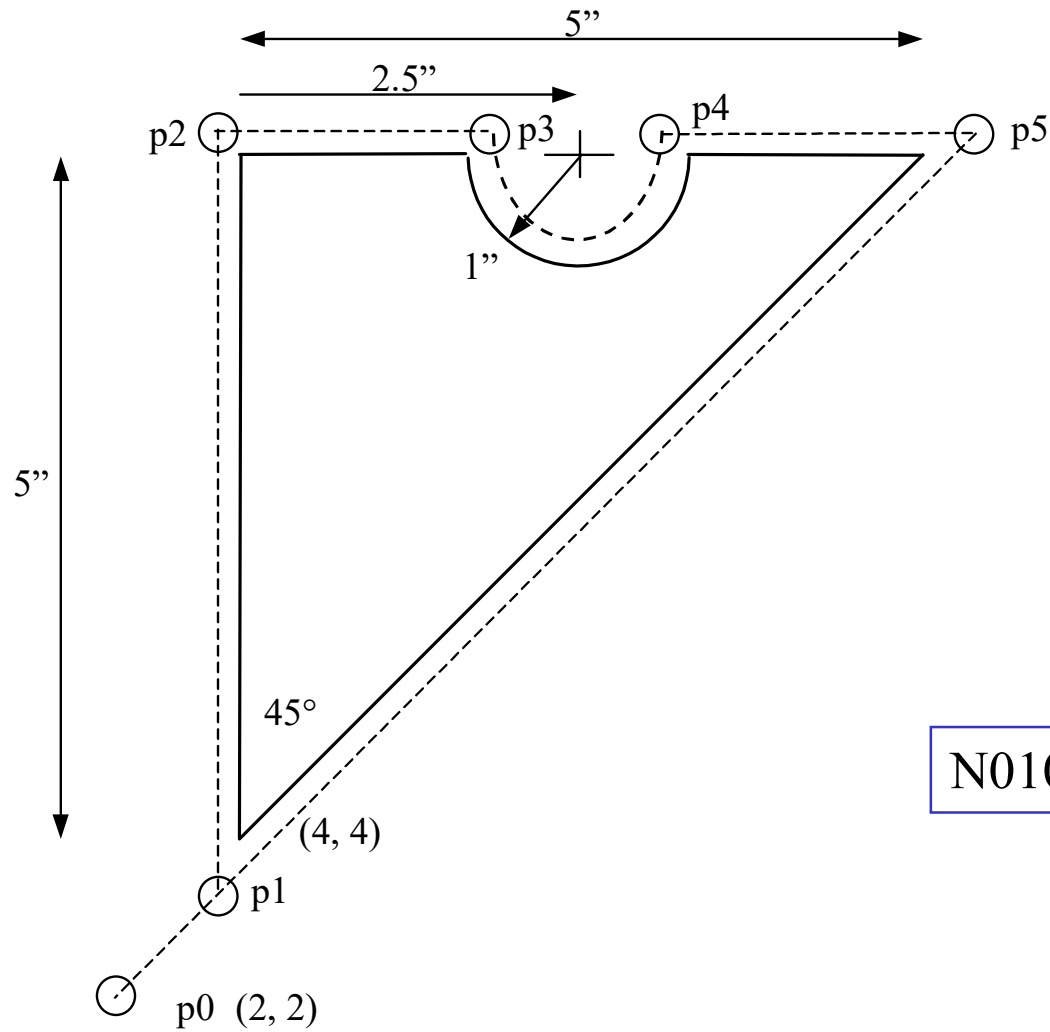


Tool size = 0.25 inch,
Feed rate = 6 inch per minute,
Cutting speed = 300 rpm,
Tool start position: 2.0, 2.0
Programming in inches

Motion of tool:

p0 → p1 → p2 → p3 → p4 → p5 → p1 → p0

1. Set up the programming parameters



Programming in inches

Use absolute coordinates

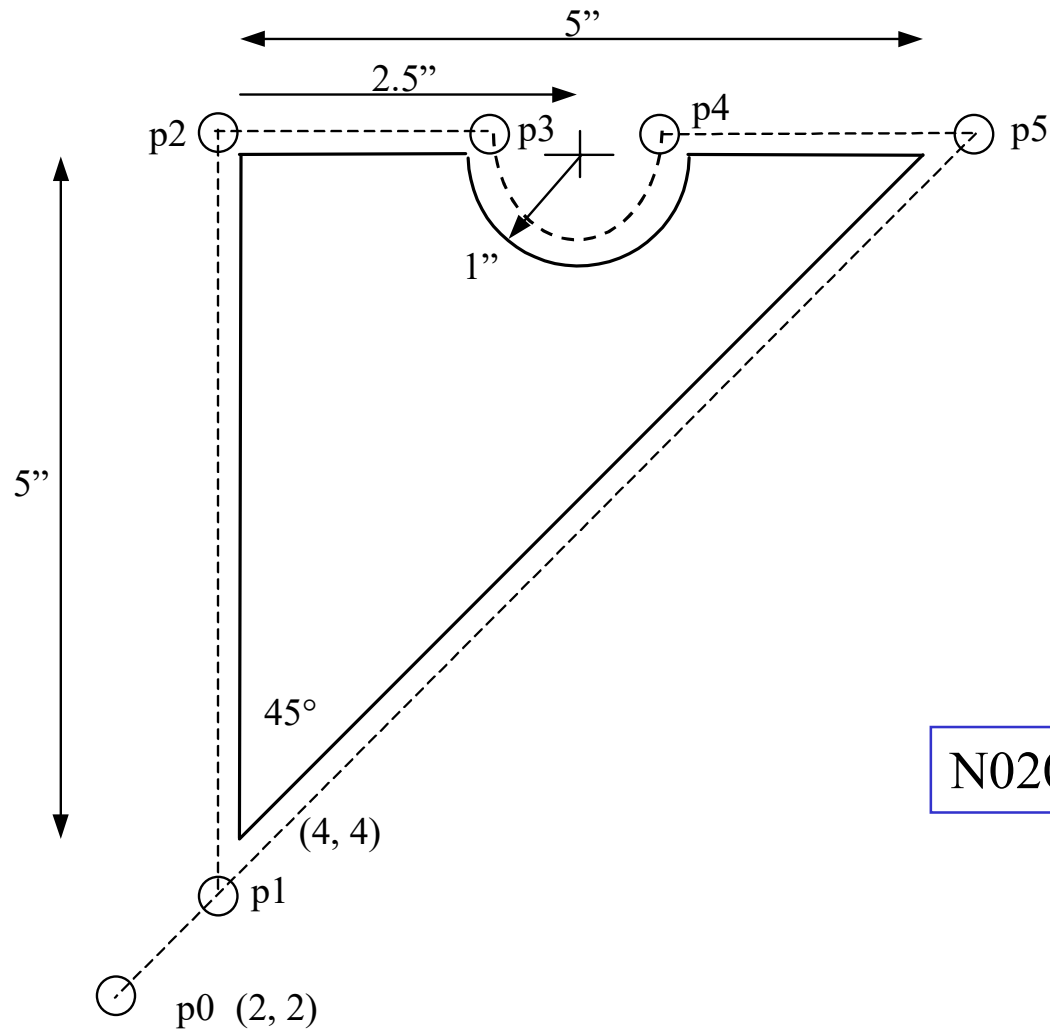
Feed in ipm

N010 G70 G90 G94 G97 M04

Spindle speed in rpm

Spindle CCW

2. Set up the machining conditions

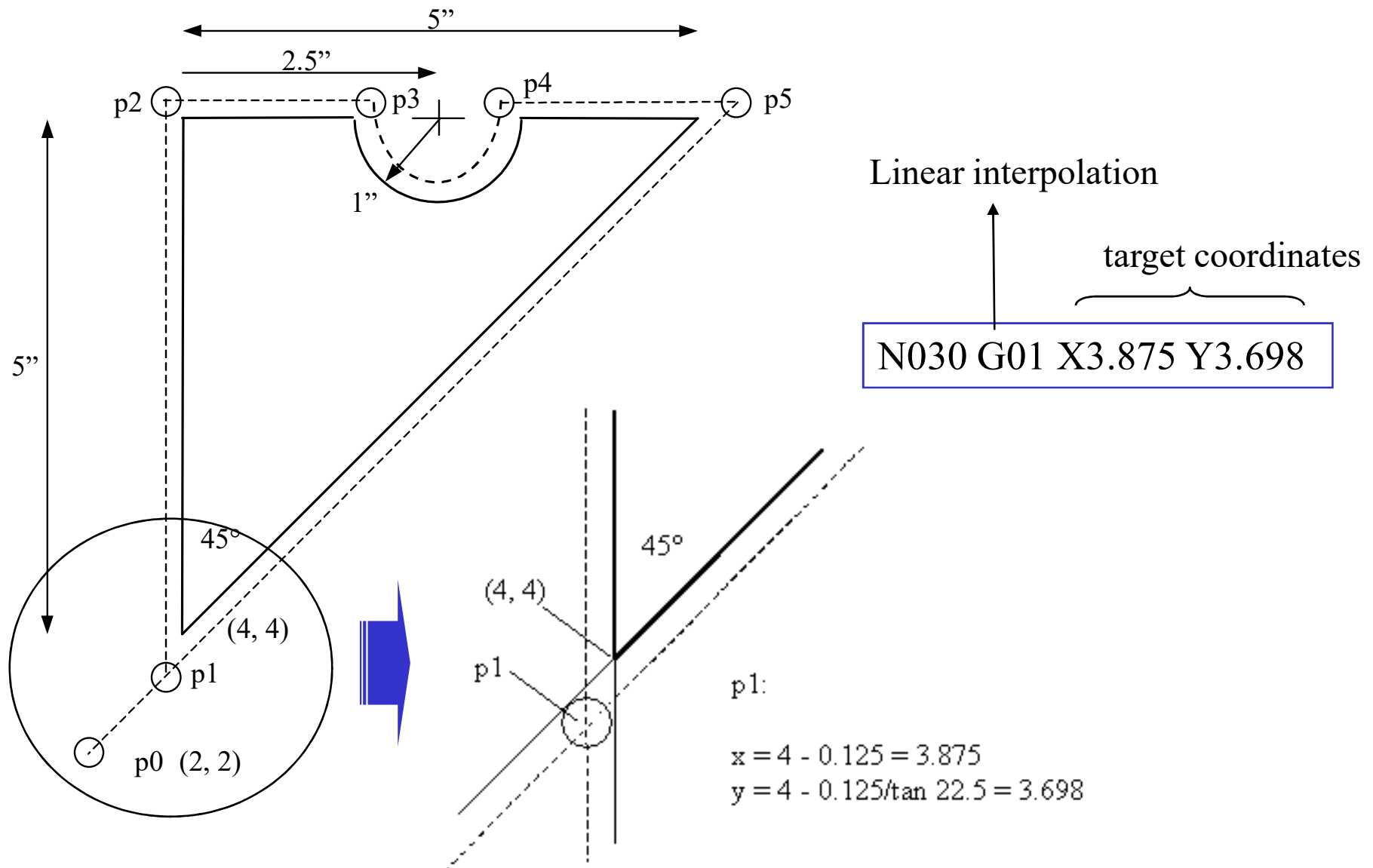


Machine moves in XY-plane
Use full-circle interpolation
Feed rate
Spindle speed

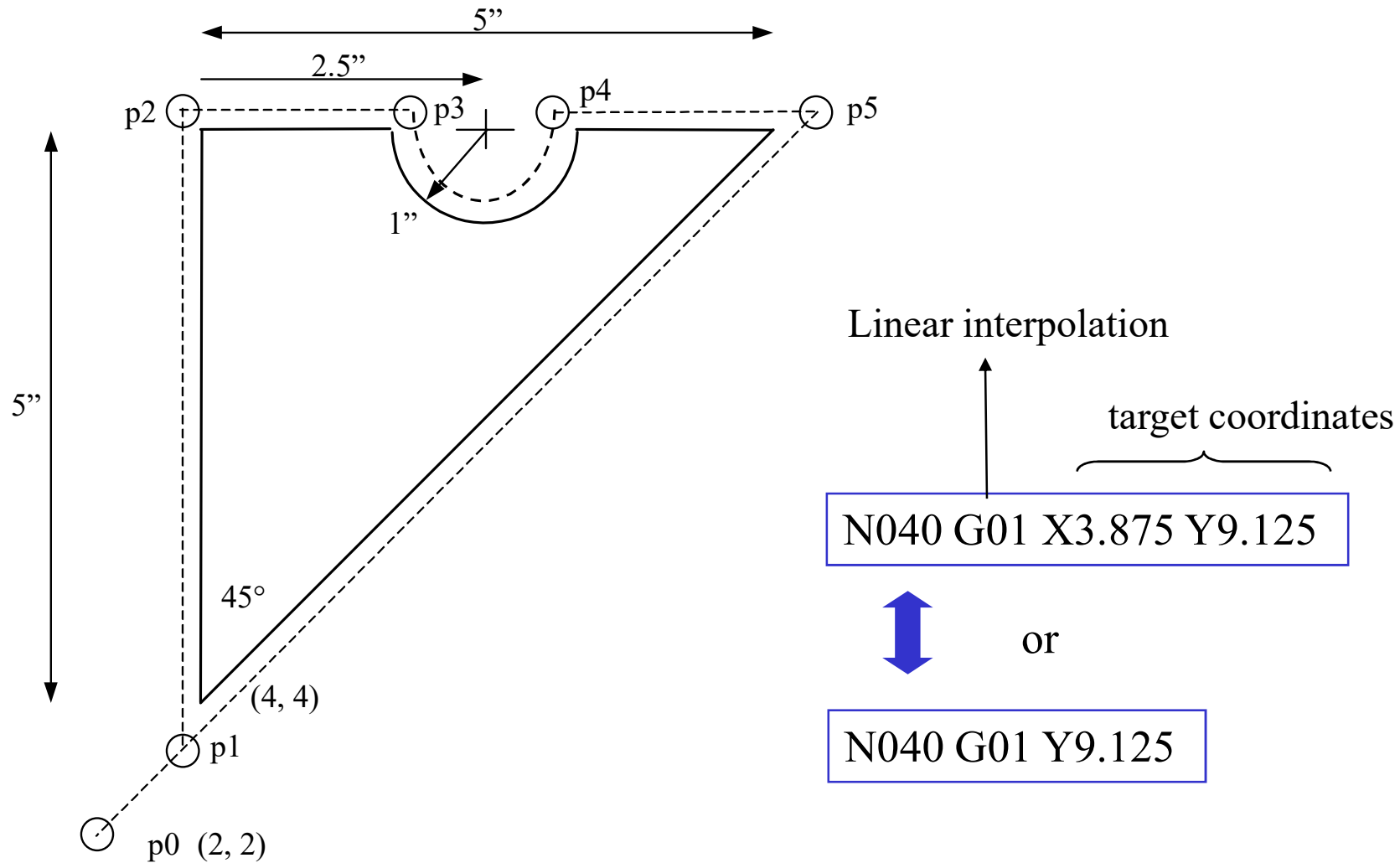
N020 G17 G75 F6.0 S300 T1001 M08

Tool no.
Flood coolant ON

3. Move tool from p0 to p1 in straight line

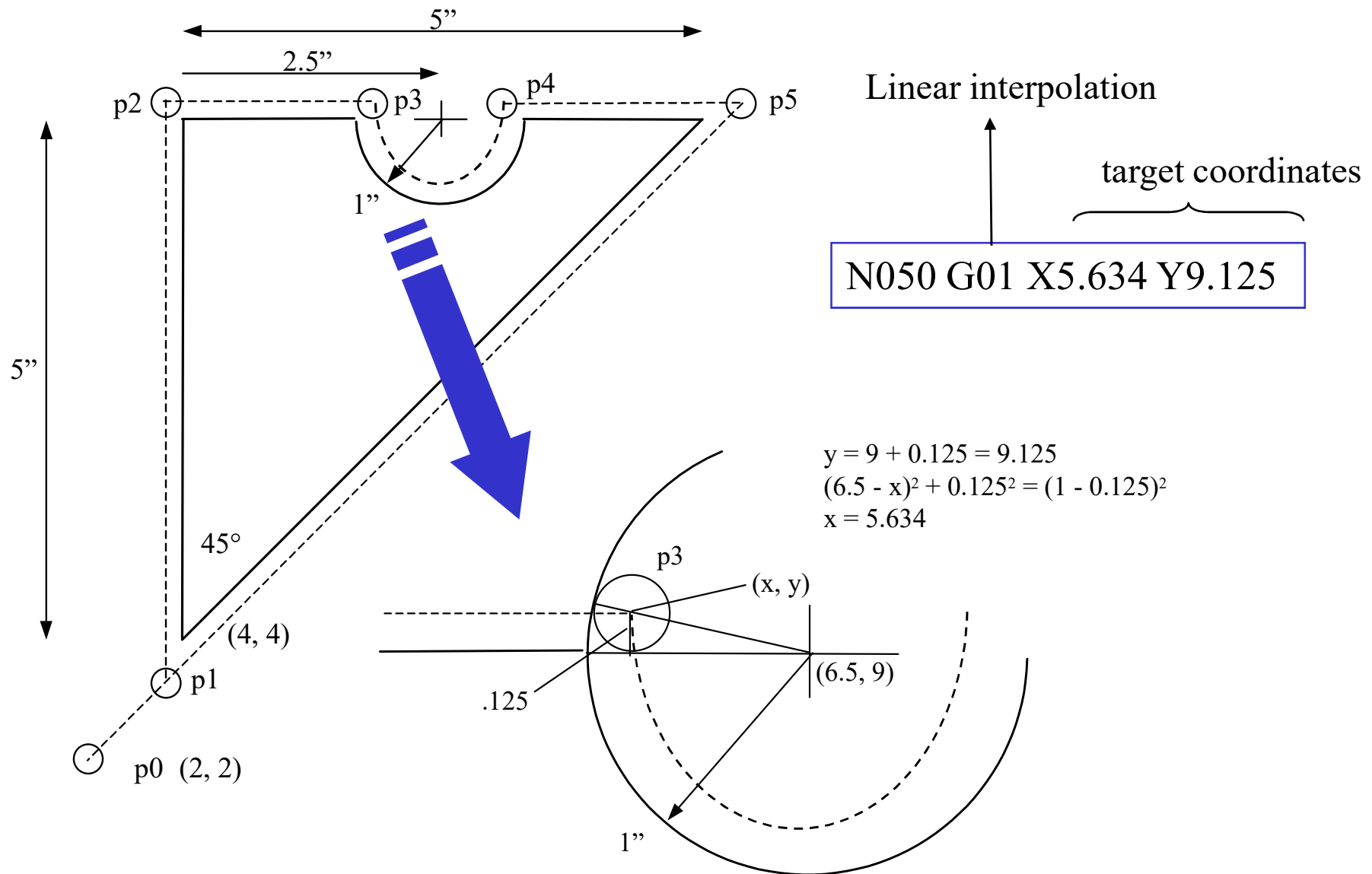


4. Cut profile from p1 to p2

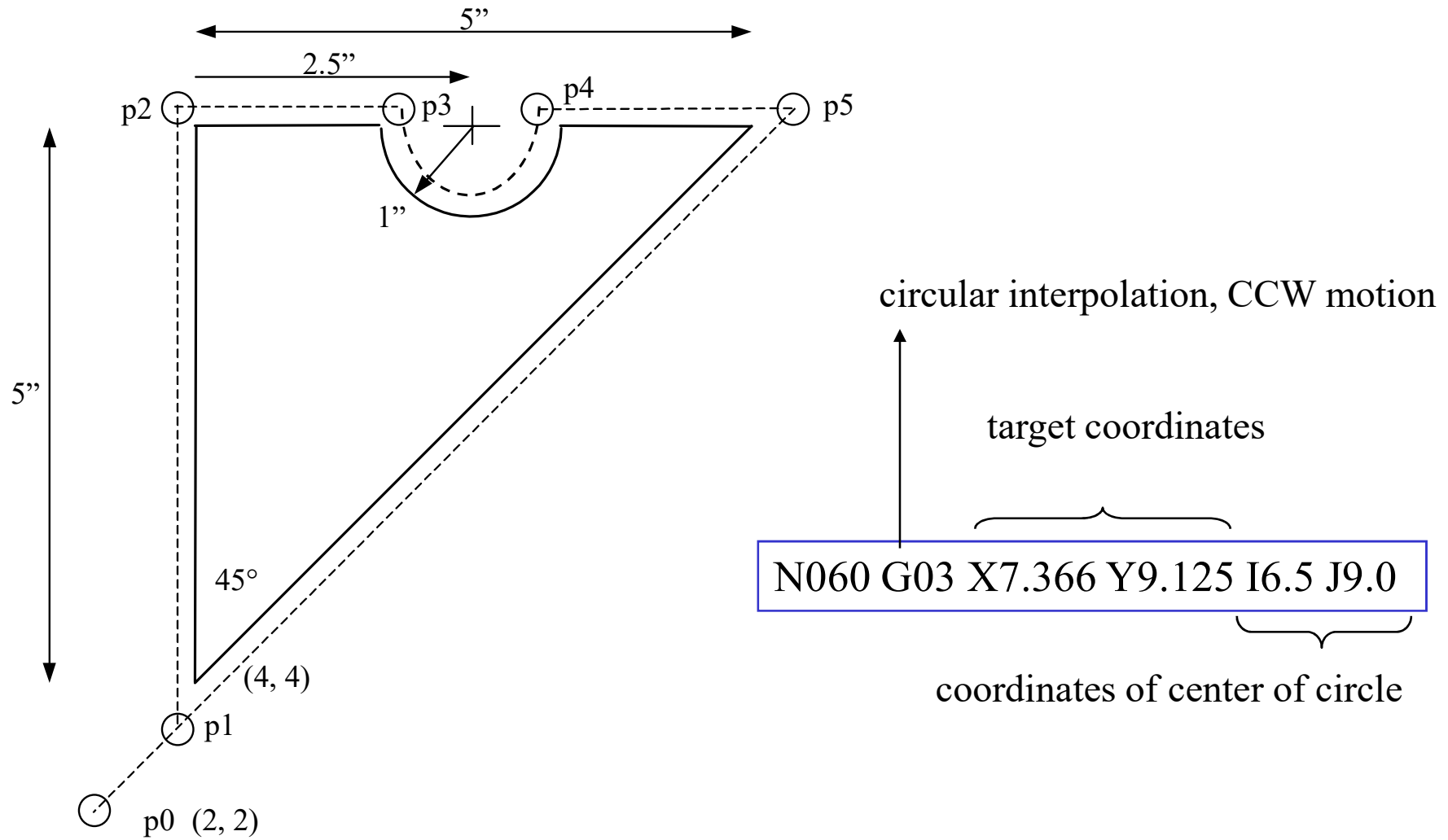


X-coordinate does not change → no need to program it

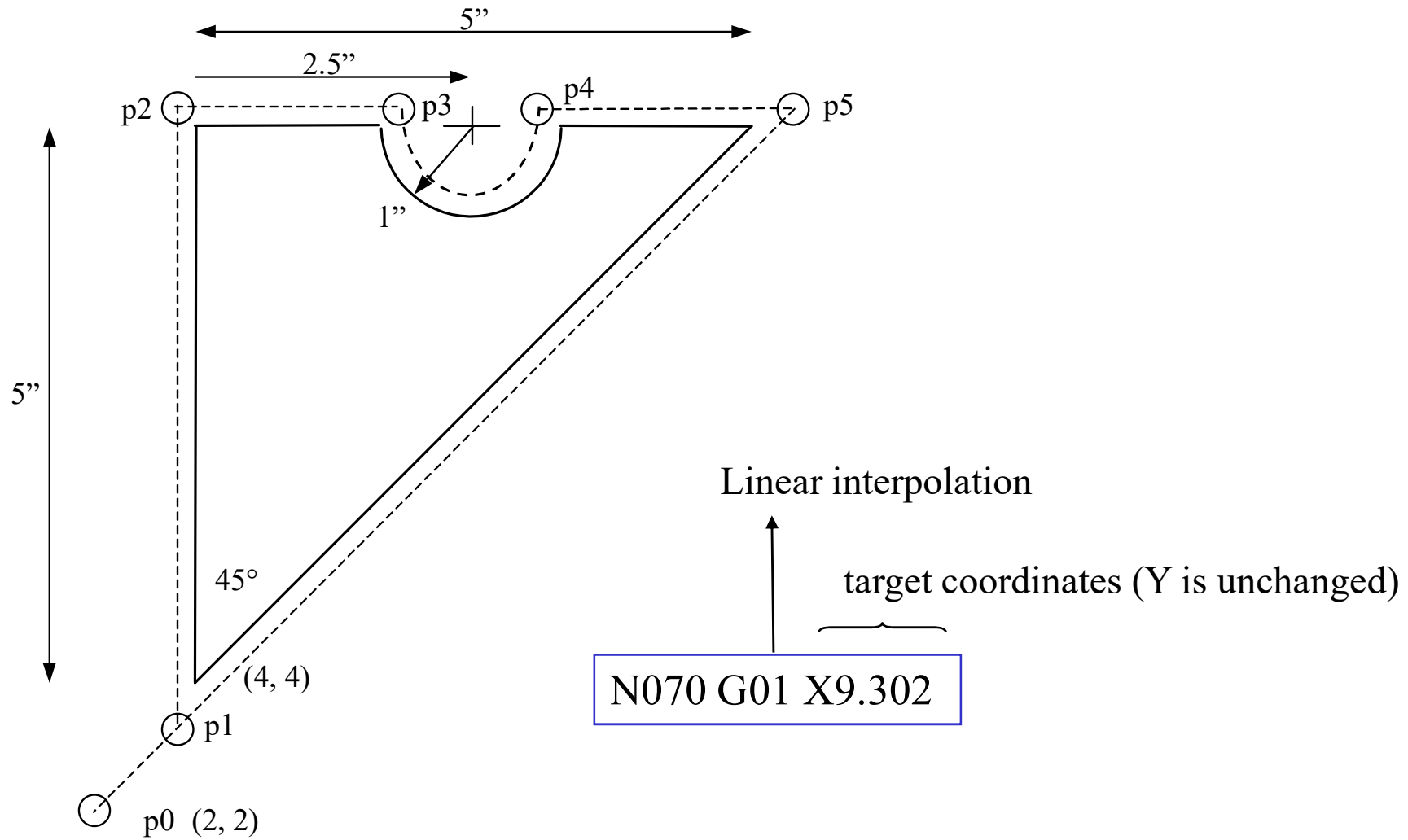
5. Cut profile from p2 to p3



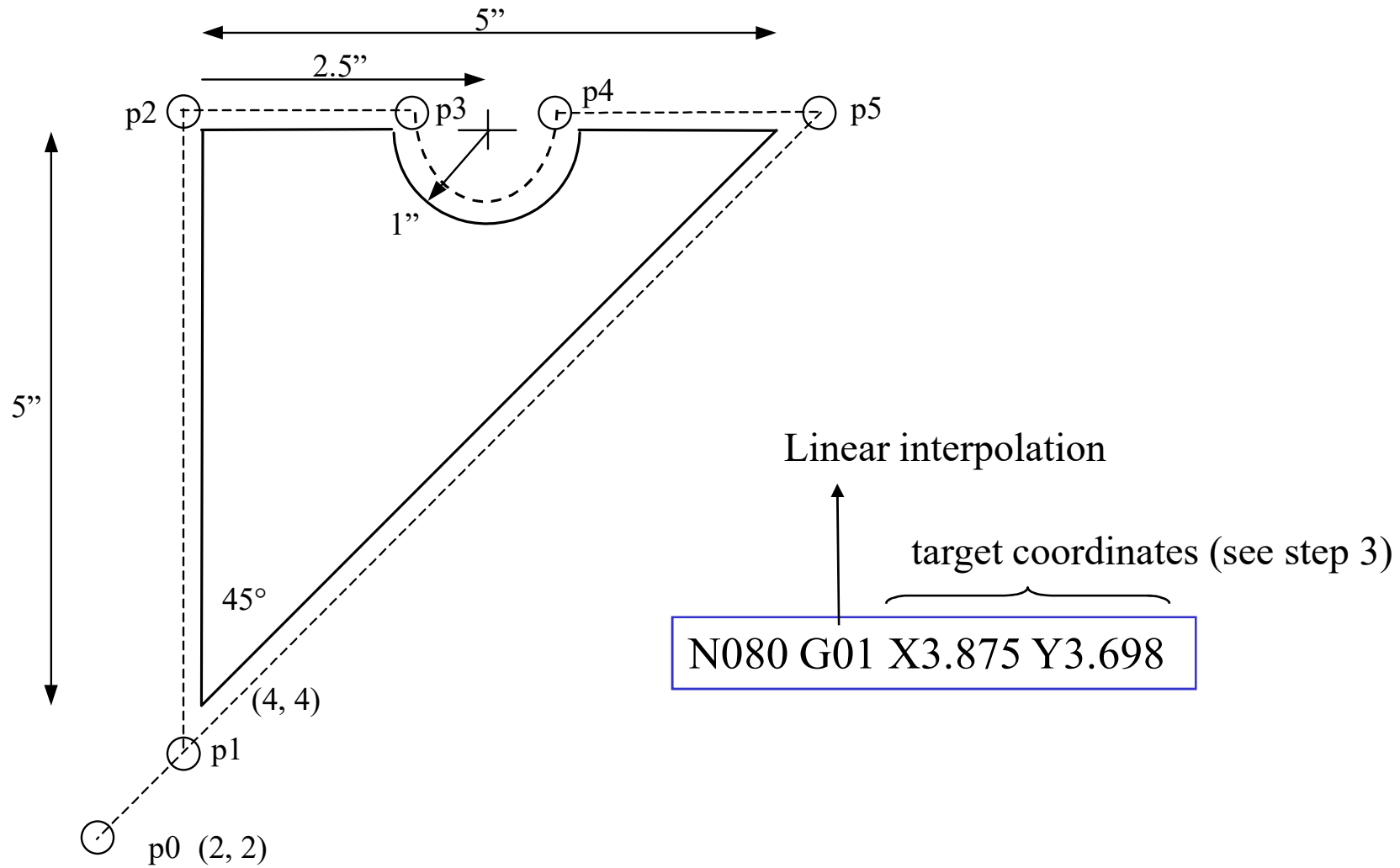
6. Cut along circle from p3 to p4



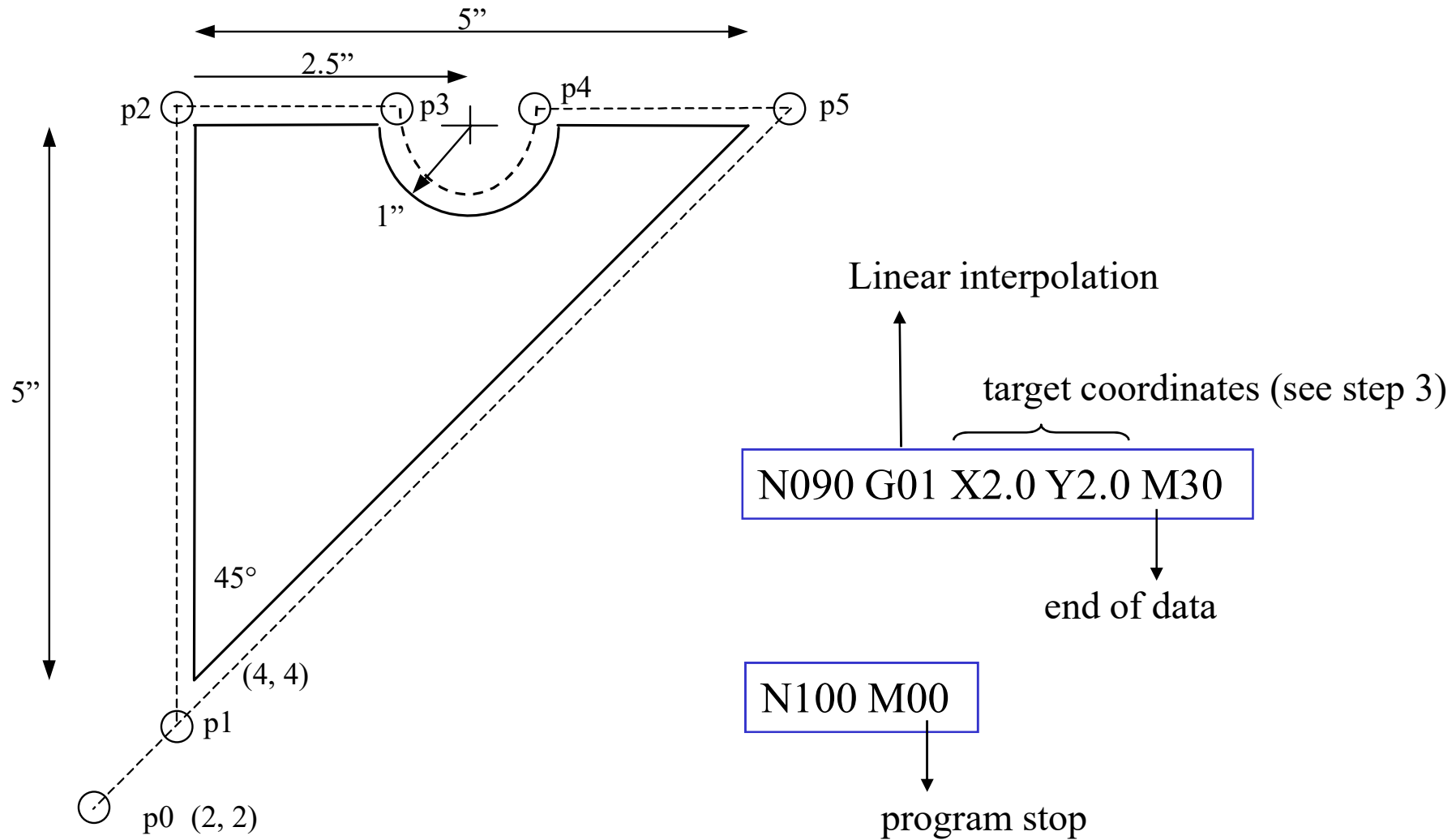
7. Cut from p4 to p5



8. Cut from p5 to p1

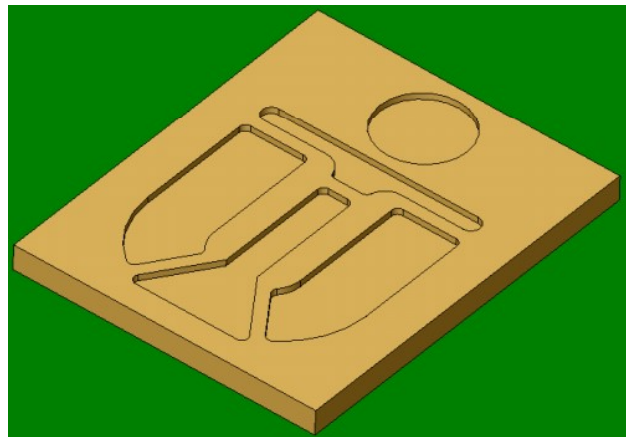


9. Return to home position, stop program

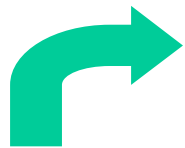


Automatic Part Programming

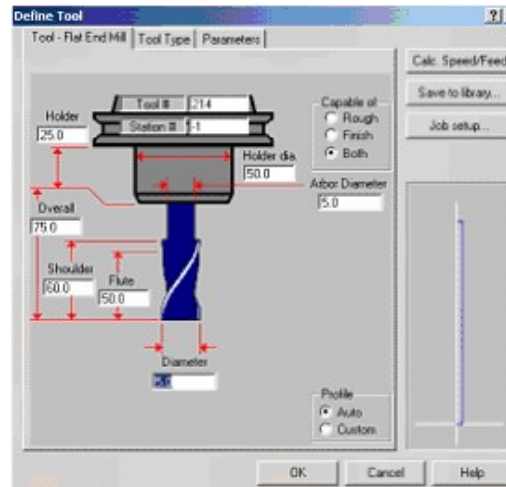
Software programs can automatic generation of CNC data



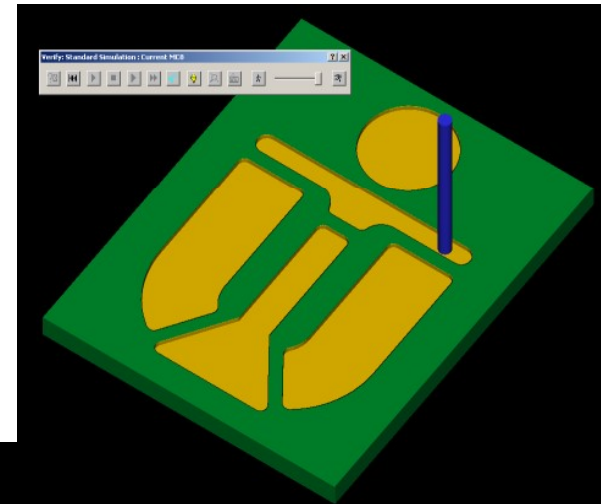
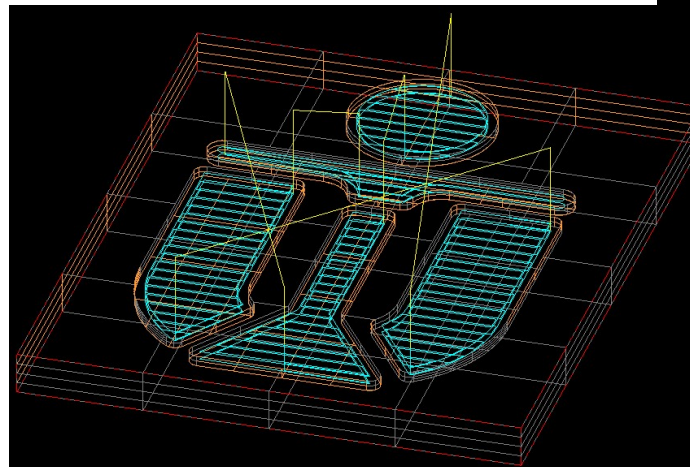
Make 3D model



Define Tool



CNC data

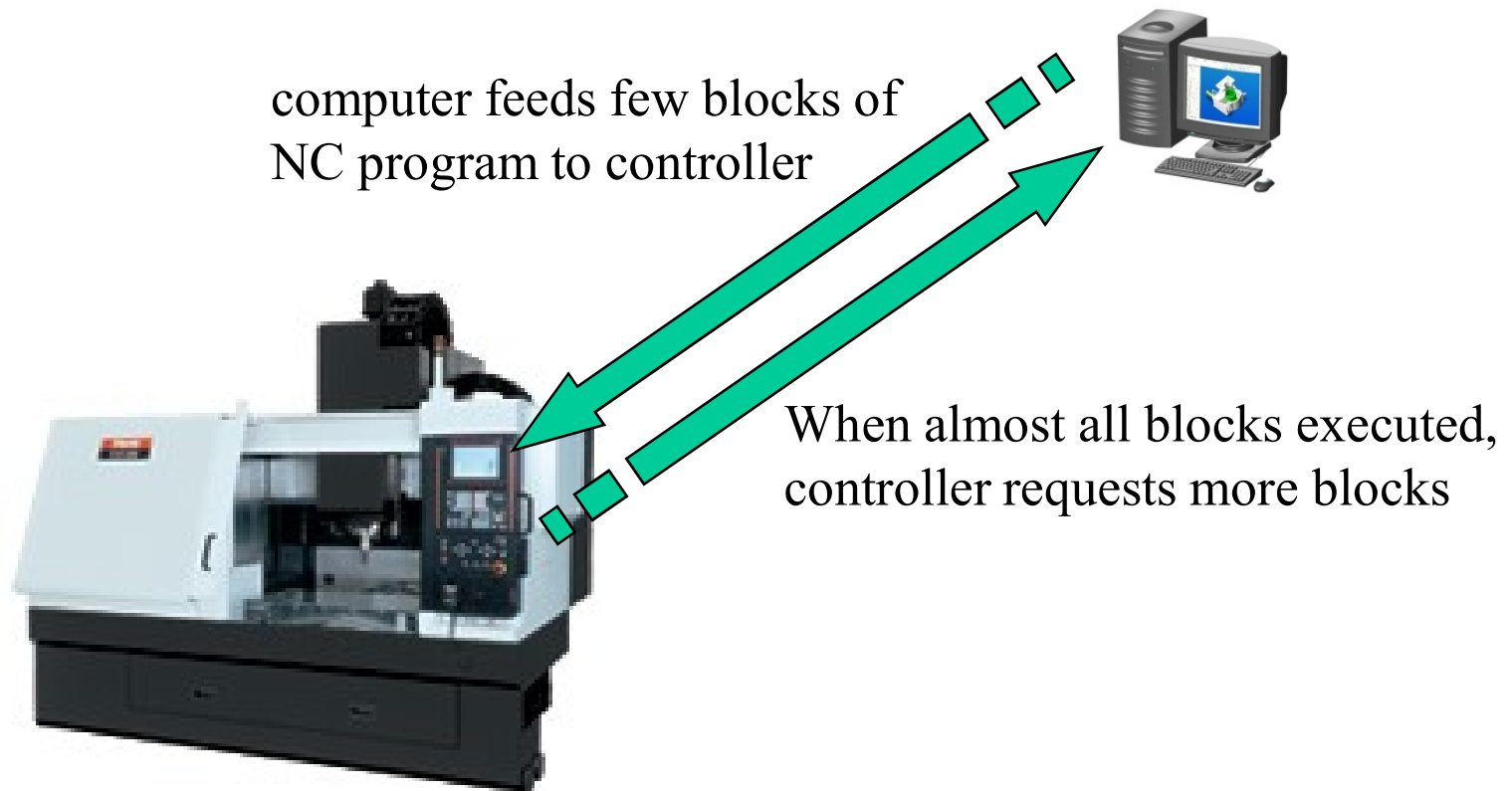


Simulate cutting

Automatic part programming and DNC

Very complex part shapes → very large NC program

NC controller memory may not handle HUGE part program



Summary

CNC machines allow precise and repeatable control in machining

CNC lathes, Milling machines, etc. are all controlled by NC programs

NC programs can be generated manually, automatically

Additional references: RS274D code descriptions